PROJECT PROPOSAL

Electricity Demand Forecasting and Spike Detection Using Spark ML and Streaming

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1. Problem

Definition. Electricity demand changes significantly throughout the day because of time-of-day patterns, weekdays/weekends, seasonality and renewable generation. This project will build a scalable system that forecasts electricity demand at 15-minute resolution, produces a complete day-ahead demand profile, and detects periods of unusually high demand or demand spikes.

Problem motivation. Forecast errors and sudden demand increases can make grid management more difficult. Using historical demand, load forecasts, solar generation, wind generation and time-based features, the system will provide demand predictions and early spike alerts.

Design Goals and Features Supported

- Day-ahead forecasting of the next day's 15-minute electricity demand
- Short-term 15-minute-ahead demand prediction for streaming monitoring.
- Use of solar and wind generation as explanatory features.
- Demand-spike detection based on unusually high demand or forecast deviation.
- Scalable batch processing with Apache Spark and Spark SQL.
- Machine-learning pipelines using Spark ML.
- Streaming simulation with Spark Structured Streaming.
- Dashboard showing actual demand, forecasts, generation features and alerts.

2. Approach

High-level design. Historical electricity-demand and generation data will be processed using Spark. Feature engineering will create lagged demand, rolling statistics, calendar variables, load-forecast error, solar generation and wind-generation features. Spark ML will be used for demand forecasting and spike classification. Historical records will then be replayed through Kafka to simulate incoming observations, which will be processed using Spark Structured Streaming and scored by the trained models.

Big Data platforms Used. Apache Spark/Spark SQL for distributed cleaning, transformation, aggregation and feature engineering; Spark ML for regression/classification; Kafka for streaming electricity observations; Spark Structured Streaming for real-time processing; and a lightweight dashboard for visualization.

Data sources and data models.

Source	Data used	Role
Open Power System Data (OPSD) 15-minute time series	Electricity load, day-ahead load forecast, wind/solar generation and related power-system variables.	Primary electricity demand and generation dataset.
Apache Spark ML documentation	Spark ML regression/classification pipeline components and evaluation APIs.	Implementation reference for scalable machine learning.

Data will be modeled as one row per 15-minute interval, with timestamp, actual electricity load, load forecast, solar generation, wind generation, calendar features and engineered lag/rolling features.

3. Evaluation Approach

Experiment plan

• Establish a seasonal/time-series baseline using previous-day and previous-week demand.
• Train a Spark ML regression model for next-15-minute demand and generate a 96-point day-ahead forecast.
• Compare models using demand history alone versus demand plus renewable-generation features.
• Define demand-spike events from large deviations from an expected time-aware baseline and evaluate a Spark ML classifier.
• Replay held-out records through Kafka at controlled rates and measure end-to-end streaming latency.
• Evaluate the final system on an unseen chronological test period.

Performance metrics for success

Forecasting	MAE, RMSE, R^2; compare against naive previous-day/previous-week baselines.
Spike detection	Precision, Recall, F1-score and confusion matrix.
Streaming	End-to-end latency, throughput (records/sec), processing lag and stability under increased event rate.
Scalability	Spark processing time and speed-up when increasing input volume/partitions.

Feature metrics for success

The system will be considered functionally successful if it can:

(1) produce 96 quarter-hour forecasts for the next day; (2) update a 15-minute-ahead prediction as new streaming observations arrive; (3) incorporate solar and wind generation into the feature set; (4) identify and alert on predefined high-demand/spike conditions; (5) display actual versus predicted demand and key generation features in the dashboard; and (6) retain traceability from each alert to the timestamp and input features that produced it.

4. Timeline / Milestones (8 Weeks)

Week	Milestone / Deliverable
1	Finalize dataset, schema, target variables and data-quality checks.
2	Build Spark ingestion/cleaning pipeline; convert data to Parquet; create 15-minute analytical tables.
3	Feature engineering and exploratory analysis; establish previous-day/previous-week baselines.
4	Develop and evaluate Spark ML demand-forecasting model.
5	Develop demand-spike detection model and finalize evaluation datasets.
6	Implement Kafka producer and Spark Structured Streaming; connect trained models for streaming inference.
7	Build dashboard; conduct end-to-end forecasting, classification and streaming experiments.
8	Final testing, scalability experiments, documentation, report and presentation/demo preparation.

Primary data reference: Open Power System Data (OPSD) Time Series package, which provides 15-minute electricity load, load forecasts, wind/solar generation and related power-system variables.